

COMP2211 Exploring AI
Fall 2025-26 Midterm Exam
28/10/2025
Time Limit: 120 Minutes

Name: _____

Stu ID: _____

Instructions:

1. This exam contains 24 pages (including this cover page) and 8 questions.
2. This is a closed-book, closed-notes examination.
3. Please sign the honor code statement on page 2.
4. Please write your answers only within the space in front pages. If you write your answers on the back of a page, please make explicit indications in front page to refer to your answers.
5. The last two pages are draft paper. It may be detached from the booklet if you wish. Please do not write your answers there.
6. You can use either pen or pencil. Please make sure your writings can be clearly read. If you use pencil, your appeal during paper checking might get rejected because of this.
7. All programming codes in your answers must be written in the Python version as taught in the class.
8. For programming questions, unless otherwise stated, you are **NOT** allowed to define additional classes, helper functions and use global variables, nor other library functions.

Grade Table (for instructor & TA use only)

Question	Points	Score
1	10	
2	20	
3	15	
4	10	
5	15	
6	10	
7	10	
8	10	
Total:	100	

The HKUST Academic Honor Code

Honesty and integrity are central to the academic work of HKUST. Students of the University must observe and uphold the highest standards of academic integrity and honesty in all the work they do throughout their program of study.

As members of the University community, students have the responsibility to help maintain the academic reputation of HKUST in its academic endeavors.

Sanctions will be imposed if students are found to have violated the regulations governing academic integrity and honesty.

Declaration of Academic Integrity

I confirm that I have answered the questions using only materials specifically approved for use in this examination, that all the answers are my own work, and that I have not received any assistance during the examination.

Signature: _____

Question 1 [10 points]

Indicate whether the following statements are true or false by putting **T** or **F** in the given table. You get 1 point for each correct answer.

Solution:

Question	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Answer	F	T	T	F	F	F	T	F	F	T

Question 2 [20 points]

(Advanced Python Programming)

1. (a) (1 point)

Solution: [7 10 13 16 19]**Marking scheme:**

1 point for fully correct answer. No partial credit.

(b) (1 point)

Solution: [] / Empty array**Marking scheme:**

1 point for fully correct answer. No partial credit.

(c) (1 point)

Solution:

[[6 6 6]

[14 14 14]]

Marking scheme:

1 point for fully correct answer. No partial credit.

(d) (2 points)

Solution:

[[[0 12]

[1 13]

[2 14]

[3 15]]

[[4 16]

[5 17]

[6 18]

[7 19]]

[[8 20]

[9 21]

[10 22]

[11 23]]]]

Marking scheme:

2 points for fully correct answer. No partial credit.

(e) (1 point)

Solution: Error.**Marking scheme:**

1 point for fully correct answer. No partial credit.

(f) (1 point)

Solution:

```
[[[ 2 3 4 5]
 [ 6 7 8 9]
 [10 11 12 13]
 [[14 15 16 17]
 [18 19 20 21]
 [22 23 24 25]]]
```

Marking scheme:

1 point for fully correct answer. No partial credit.

Acceptable deviation:

- Answers put in `np.array([...])`
- Different alignment as long as `[]` are put correctly

Extra/missing outmost `[]`: -1 once for the whole question.

2. (a) (1 point)

Solution:

```
game_reviews.transpose(0, 2, 1) /  
game_reviews.transpose((0, 2, 1)) /  
np.transpose(game_reviews, (0, 2, 1))
```

Marking scheme:

1 point for fully correct answer. No partial credit.

(b) (1 point) Write the output of the following Python statement:

```
# np.shares_memory(a, b): determine if two arrays share memory.  
print(np.shares_memory(game_reviews, transposed_reviews))
```

Solution: True / 1

Marking scheme:

1 point for fully correct answer. No partial credit.

(c) (1 point)

Solution:

```
game_reviews[ [0, 2] , 1 , ::2 ]
```

Other possible solutions:

- 1st dimension: ::2
- 2nd dimension: 1: / [1]
- 3rd dimension: 0:7:2

Marking scheme:

1 point for fully correct answer. No partial credit.

(d) (1 point)

Solution:

```
game_reviews[ : , [[0], [1]] , [[0, 1], [1, 2]] ] /  
game_reviews[ : , [[0, 0], [1, 1]] , [[0, 1], [1, 2]] ]
```

Marking scheme:

1 point for fully correct answer. No partial credit.

(e) (2 points) Write one line of Python code to extract the review sentiment (True if the number of positive reviews is greater than or equal to the number of

negative reviews; False otherwise) for each game on each day, resulting in a 3×7 Boolean array.

`review_sentiment =`

Solution:

```
game_reviews[:, 0, :] - game_reviews[:, 1, :] >= 0 /
game_reviews[:, 0] - game_reviews[:, 1] >= 0 /
game_reviews[:, 0, :] >= game_reviews[:, 1, :]
```

Marking scheme:

- 2 point for fully correct answer.
- 1 point for minor mistake, e.g., `>=` becomes `>` / `<` / `==`,
: becomes `...`, extracting the values from `game_reviews`.
- No further partial credit.

(f) (2 points)

Solution:

```
game_reviews.sum(axis=1) >= 150 /
game_reviews.sum(1) >= 150
np.sum(game_reviews, axis=1) >= 150 /
game_reviews[:, 0] + game_reviews[:, 1] >= 150
```

Marking scheme:

- 2 point for fully correct answer.
- 1 point for minor mistake, e.g., `>=` becomes `>` / `<` / `==`,
`np.game_reviews`, extracting the values from `game_reviews`.
- No further partial credit.

(g) (2 points)

Solution:

```
game_reviews[2, 1, :][filtered_mask[2]].min() /
game_reviews[2, 1, filtered_mask[2]].min() /
game_reviews[2][1][filtered_mask[2]].min() /
np.min(game_reviews[2, 1, :][filtered_mask[2]])
```

Marking scheme:

- 2 point for fully correct answer.
- 1 point for minor mistake, e.g., missing ().
- No further partial credit.

(h) (1 point)

Solution:

```
game_reviews[:, 0, :].sum(axis=1) /  
game_reviews[:, 0].sum(1) /  
game_reviews.sum(2)[:, 0]
```

Marking scheme:

1 point for fully correct answer. No partial credit.

(i) (2 points)

Solution:

```
np.abs(num_positive_reviews[:, None] -  
      num_positive_reviews[None, :])
```

Other possible methods for expanding dimension:

- `num_positive_reviews.reshape(3,1)`
- `np.expand_dims(num_positive_reviews)`
- `num_positive_reviews[:, np.newaxis]`

Marking scheme:

- 2 point for fully correct answer.
- 1 point for minor mistake, e.g., not absolute value, `[:, :, None]`, wrong variable name.
- No further partial credit.

Question 3 [15 points]
(Naïve Bayes Model)

1. (a) (1 point)

Solution:

$$P(\text{Finish_HW} = \text{Yes}) = 0.6,$$

$$P(\text{Finish_HW} = \text{No}) = 0.4.$$

Marking scheme:

- 0.5 points for each correct propr probability. 1 point in total.
- No further partial credit.

(b) (2 points)

Solution:

$$P(\text{Weather} = \text{Sunny} \mid \text{Yes}) = \frac{2}{6}, P(\text{Weather} = \text{Sunny} \mid \text{No}) = \frac{2}{4}.$$

$$P(\text{Got_Reminder} = \text{True} \mid \text{Yes}) = \frac{4}{6}, P(\text{Got_Reminder} = \text{True} \mid \text{No}) = \frac{1}{4}.$$

Marking scheme:

- 0.5 points for each correct likelihood. 2 points in total.
- No further partial credit.

(c) (3 points)

Solution:

$\begin{aligned} \sigma &= \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2} \\ \mu_{\text{Yes}} &= 3.75 \\ \sigma_{\text{Yes}} &= \sqrt{7/8} \approx 0.9354 \\ \mu_{\text{No}} &= 1.75 \\ \sigma_{\text{No}} &= \sqrt{\frac{5}{12}} \approx 0.6454 \\ p(x = 3 \mid \mu_{\text{Yes}}, \sigma_{\text{Yes}}) &\approx 0.31 \\ p(x = 3 \mid \mu_{\text{No}}, \sigma_{\text{No}}) &\approx 0.09 \end{aligned}$	$\begin{aligned} \sigma &= \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \\ \mu_{\text{Yes}} &= 3.75 \\ \sigma_{\text{Yes}} &= \sqrt{70/96} \approx 0.85 \\ \mu_{\text{No}} &= 1.75 \\ \sigma_{\text{No}} &= \sqrt{\frac{5}{16}} \approx 0.56 \\ p(x = 3 \mid \mu_{\text{Yes}}, \sigma_{\text{Yes}}) &\approx 0.32 \\ p(x = 3 \mid \mu_{\text{No}}, \sigma_{\text{No}}) &\approx 0.05899 \end{aligned}$
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Marking scheme:

- 0.5 points for each correct term. 3 points in total.
- No further partial credit.

(d) (3 points)

Solution:

$$P(\text{Yes}|X_{\text{new}}) = \frac{P(\text{Yes}) \times P(\text{Sunny}|\text{Yes}) \times P(\text{True}|\text{Yes}) \times p(x = 3.0|\text{Yes})}{P(X_{\text{new}})}$$

$$P(\text{No}|X_{\text{new}}) = \frac{P(\text{No}) \times P(\text{Sunny}|\text{No}) \times P(\text{True}|\text{No}) \times p(x = 3.0|\text{No})}{P(X_{\text{new}})}$$

$$\left| \begin{array}{l} \sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2} \\ P(\text{Yes}|X_{\text{new}}) = \frac{0.041}{0.0455} \approx 0.90 \\ P(\text{No}|X_{\text{new}}) = \frac{0.0045}{0.0455} \approx 0.10 \end{array} \right| \left| \begin{array}{l} \sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \\ P(\text{Yes}|X_{\text{new}}) = \frac{0.0427}{0.0457} \approx 0.934 \\ P(\text{No}|X_{\text{new}}) = \frac{0.003}{0.0457} \approx 0.0656 \end{array} \right|$$

We have $P(\text{Yes}|X_{\text{new}}) > P(\text{No}|X_{\text{new}})$, therefore X_{new} belongs to class Yes.

Marking scheme:

- 1 point for correct posterior probability calculation formula (only consider numerator), 0.5 point for each.
- 1 point for correct posterior probability calculation result (only consider numerator), 0.5 point for each.
- 1 point for correct prediction.
- No further partial credit.

2. (a) (3 points)

Solution:

$$P(\text{Finish_HW}) \cdot P(\text{Weather} = \text{Sunny}, \text{Got_Reminder} = \text{True} | \text{Finish_HW}) \\ \cdot p(x = 3.0 | \text{Finish_HW}) / P(\text{Weather} = \text{Sunny}, \text{Got_Reminder} = \text{True}, \text{Study_Hours} = 3.0)$$

Marking scheme:

- 3 points for fully correct formula: (Sunny, True, $x = 3$ can be omitted)
 - 0.5 points for correct prior probability $P(\text{Finish_HW})$
 - 1 point for correct joint likelihood $P(\text{Weather} = \text{Sunny}, \text{Got_Reminder} = \text{True} | \text{Finish_HW})$.
 - 0.5 points for correct likelihood $p(x = 3.0 | \text{Finish_HW})$
 - 0.5 points for correct marginal probability $P(\text{Weather} = \text{Sunny}, \text{Got_Reminder} = \text{True}, \text{Study_Hours} = 3.0)$
 - 0.5 points for correct combination of these terms.

(b) (3 points)

Solution: First compute the joint likelihoods,

$$P(\text{Weather} = \text{Sunny}, \text{Got_Reminder} = \text{True} | \text{Finish_HW} = \text{Yes}) = \frac{2}{6},$$

$$P(\text{Weather} = \text{Sunny}, \text{Got_Reminder} = \text{True} | \text{Finish_HW} = \text{No}) = \frac{1}{4},$$

Then we have the posteriors,

$$\left| \begin{array}{l} \sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2} \\ P(\text{Yes} | X_{\text{new}}) = \frac{0.062}{X_{\text{new}}} \\ P(\text{No} | X_{\text{new}}) = \frac{0.009}{X_{\text{new}}} \end{array} \right| \left| \begin{array}{l} \sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \\ P(\text{Yes} | X_{\text{new}}) = \frac{0.064}{X_{\text{new}}} \\ P(\text{No} | X_{\text{new}}) = \frac{0.006}{X_{\text{new}}} \end{array} \right|$$

The marginal probability $P(X_{\text{new}})$ doesn't influence the prediction. X_{new} belongs to class Yes.

Marking scheme:

- 0.5 points for each correct joint likelihood. 1 point in total. (If they calculate correct posterior probability, they get the points.)
- 0.5 points for each correct posterior probability (no need to give the value of $P(X_{\text{new}})$). 1 point in total.
- 1 point for correct prediction.
- No further partial credit.

Question 4 [10 points]
(K-Nearest Neighbors)

1. (2 points)

Solution: (Only one is enough) (1.5, 64.9), (1.5, 35.1) or (1.5, 85.6)

Marking scheme:

2 points for correct answer. No partial credit.

2. (1 point)

Solution: (Only two are enough) Manhattan distance (*aka.* City block distance), Cosine distance, Hamming distance.

Marking scheme:

0.5 points for each correct name of distance function. 1 point in total. No further partial credit.

3. (2 points)

Solution: **Game** will always be predicted.

With $K = 5$:

- Utility: 2 (Index 1, 3)
- Game: 3 (Index 2, 4, 5)

$\Rightarrow$ Majority class = **Game** $\Rightarrow$ All new apps predicted as **Game**, regardless of their features.

KNN becomes a majority-class classifier when $K = N$.

Marking scheme:

- 1 point for correct answer of “Game”.
- 1 point for correct justification.
- No further partial credit.

4. (5 points)

Solution: Step 1: Normalization setup.

Training-set ranges: $x_1 \in [1, 5] \Rightarrow \Delta x_1 = 4$, $x_2 \in [20, 110] \Rightarrow \Delta x_2 = 90$. Thus $x'_1 = (x_1 - 1)/4$, $x'_2 = (x_2 - 20)/90$.

Normalized training points:

Idx 1 (1, 20) $\rightarrow$ (0, 0) [Utility], Idx 2 (4, 50) $\rightarrow$ (0.75, 0.33 $\bar{3}$) [Game],
 Idx 3 (2, 80) $\rightarrow$ (0.25, 0.66 $\bar{6}$) [Utility], Idx 4 (5, 90) $\rightarrow$ (1, 0.77 $\bar{7}$) [Game],
 Idx 5 (3, 110) $\rightarrow$ (0.5, 1.0) [Game].

Step 2: Predict X_{test} .

X_{test} (3, 100) $\Rightarrow (x'_1, x'_2) = ((3-1)/4, (100-20)/90) = (0.5, 0.88\bar{8})$. Euclidean distances to training (all on normalized space):

$$d(X_{test}, \text{Idx 1}) = \sqrt{(0.5 - 0.0)^2 + (0.88\bar{8} - 0.0)^2} \approx 1.020 \pm 0.001,$$

$$d(X_{test}, \text{Idx 2}) = \sqrt{(0.5 - 0.75)^2 + (0.88\bar{8} - 0.33\bar{3})^2} \approx 0.610 \pm 0.001,$$

$$d(X_{test}, \text{Idx 3}) = \sqrt{(0.5 - 0.25)^2 + (0.88\bar{8} - 0.66\bar{6})^2} \approx 0.334 \pm 0.001,$$

$$d(X_{test}, \text{Idx 4}) = \sqrt{(0.5 - 1)^2 + (0.88\bar{8} - 0.77\bar{7})^2} \approx 0.512 \pm 0.001,$$

$$d(X_{test}, \text{Idx 5}) = \sqrt{(0.5 - 0.5)^2 + (0.88\bar{8} - 1)^2} \approx 0.112 \pm 0.001.$$

$K = 3$ neighbors: Idx 5 (Game), Idx 3 (Utility), Idx 4 (Game). Majority vote $\Rightarrow$ Game. Hence X_{test} 's *Predicted* label is **Game**.

Marking scheme:

- 0.5 points for correct normalized testing data point.
- 0.5 points for each correct normalized training data point. 2.5 points in total.
- 0.5 points for each correct nearest neighbor and its correct distance calculation (if present on the test paper). 1.5 points in total.
- 0.5 points for correct prediction.
- No further partial credit.

Question 5 [15 points]

(D-fold Cross Validation and Model Evaluation)

1. (1 point)

Solution: D models are trained. $D \cdot q$ errors are recorded.**Marking scheme:**

- 0.5 points for D models.
- 0.5 points for $D \cdot q$ errors.
- No further partial credit.

2. (1.25 points)

Solution: The dataset should be shuffled before dividing it into folds. This will ensure that each fold contains a representative distribution of samples from all classes, reducing bias and providing more reliable evaluation results.**Marking scheme:**

- 1.25 points for correct answer, must mention terms like “shuffle”, or “mix up”.
- 0.75 points for minor mistake, such as answering with “randomly select the samples”.
- No further partial credit.

3. (a) (0.5 points)

Solution: For each choice of K , find the average error across validation sets. Choose the value of K with the lowest average error.**Marking scheme:**

0.5 points for correct answer. No partial credit.

(b) (0.5 points)

Solution: A low K-value is sensitive to outliers and a higher K-value is more resilient to outliers as it considers more voters to decide prediction.**Marking scheme:**

- 0.25 points for stating “A low K-value is sensitive to outliers”.
- 0.25 points for stating “a higher K-value is more resilient to outliers as it considers more voters to decide prediction”.

- No further partial credit.

4. (a) (2.25 points)

Solution:

$2 \ 2 \ 0$

$1 \ 1 \ 2$

$1 \ 0 \ 3$

Marking scheme:

0.25 points for each correct entry. 2.25 points in total. No further partial credit.

(b) (3 points)

Solution:

$2 \ 6 \ 2 \ 2$

$1 \ 6 \ 2 \ 3$

$3 \ 6 \ 2 \ 1$

Marking scheme:

0.25 points for each correct entry. 3 points in total. No further partial credit.

(c) (3 points)

Solution:

$1/2 \quad 1/2 \quad 1/2 \quad 1/4$

$1/3 \quad 1/4 \quad 2/7 \ (0.286) \quad 0$

$3/5 \quad 3/4 \quad 2/3 \quad 0.478 \ (4/\sqrt{70})$

Marking scheme:

0.25 points for each correct entry. 3 points in total. Every improper decimal will get 0.1 deduction. No further partial credit.

(d) (1.5 points) According to evaluation metrics computed above, how does the KNN model perform on each class (0,1,2)?

Solution: The KNN model can make some reasonable good predictions on class 2, due to the highest Precision, Recall, F1-score and MCC; KNN model predicting class 0 is closer to a random guess classifier because of the lower MCC score it gets (0.25 is closer to 0); Since MCC for class 1 is 0, then KNN model seems to randomly guess on class 1.

Marking scheme:

0.5 points for correct analysis on each class. 1.5 points in total. No further partial credit.

- i. If the answer make correct comparison between class 0,1,2, which is class $2 > \text{class } 0 > \text{class } 1$, it is also correct;

- ii. If only general comparison is mentioned, like performance is not good, the answer can only get 0.5 points.

(e) (2 points)

Solution:

$Total\ TP = 2 + 1 + 3 = 6$; $Total\ FP = 2 + 2 + 2 = 6$; $Total\ FN = 2 + 3 + 1 = 6$;

$Accuracy = 6/12 = 1/2$;

$Micro\ F1 - score = 2 \times 6 / (2 \times 6 + 6 + 6) = 1/2$;

$Macro\ F1 - score = (1/2 + 2/7 + 2/3)/3 = 0.484\ (61/126)$;

$Weighted\ F1 - score = 0.484\ (61/126)$.

Marking scheme:

0.5 points for each correct evaluation metric. 2 points in total. Any improper decimal will be deducted 0.1 points. No further partial credit.

Question 6 [10 points]
(K-Means Clustering)

1. (1.5 points)

Solution: Distances (sketch):

- $A \rightarrow C_1^{(0)}$ (0 vs $\sqrt{53}$ vs $\sqrt{37}$) $\Rightarrow C_1$
- $B \rightarrow C_1^{(0)}$ ($\sqrt{2}$ vs $\sqrt{37}$ vs $\sqrt{29}$) $\Rightarrow C_1$
- $C \rightarrow C_2^{(0)}$ ($\sqrt{53}$ vs 0 vs $\sqrt{80}$) $\Rightarrow C_2$
- $D \rightarrow C_2^{(0)}$ ($\sqrt{52}$ vs $\sqrt{5}$ vs $\sqrt{53}$) $\Rightarrow C_2$
- $E \rightarrow C_2^{(0)}$ ($\sqrt{50}$ vs $\sqrt{13}$ vs $\sqrt{37}$) $\Rightarrow C_2$
- $F \rightarrow C_3^{(0)}$ ($\sqrt{37}$ vs $\sqrt{80}$ vs 0) $\Rightarrow C_3$

Clusters after assignment: $C_1 \leftarrow \{A, B\}$, $C_2 \leftarrow \{C, D, E\}$, $C_3 \leftarrow \{F\}$.

Marking scheme:

0.25 points for each correct clustered data point. 1.5 points in total. No further partial credit.

2. (1.5 points)

Solution:

$$\begin{aligned} C_1^{(1)} &= \left(\frac{2+3}{2}, \frac{8+7}{2} \right) = (2.5, 7.5), \\ C_2^{(1)} &= \left(\frac{4+6+7}{3}, \frac{1+2+3}{3} \right) = \left(\frac{17}{3}, 2 \right) \approx (5.67, 2), \\ C_3^{(1)} &= (8, 9). \end{aligned}$$

Marking scheme:

0.5 points for each correct centroid. 1.5 points in total. No further partial credit.

3. (2 points)

Solution:

$$\begin{aligned} \text{SSE}_{C_1} &= \|(2, 8) - (2.5, 7.5)\|^2 + \|(3, 7) - (2.5, 7.5)\|^2 = 0.5 + 0.5 = 1, \\ \text{SSE}_{C_2} &= \|(4, 1) - (5.67, 2)\|^2 + \|(6, 2) - (5.67, 2)\|^2 + \|(7, 3) - (5.67, 2)\|^2 \\ &= \frac{34}{9} + \frac{1}{9} + \frac{25}{9} = \frac{60}{9} = \frac{20}{3}, \\ \text{SSE}_{C_3} &= 0 \quad (\text{singleton}). \end{aligned}$$

Thus $SSE = 1 + \frac{20}{3} = \boxed{\frac{23}{3} \approx 7.67}$.

Marking scheme:

- 2 points for correct final answer.
- If the final answer is not correct, 0.5 points for each correct cluster's SSE.
- No further partial credit.

4. (a) (1.5 point)

Solution: A centroid (C_3) will be pulled toward G , inflating SSE. Others (C_1 and C_2) would change little/no change in the first iteration.

Marking scheme:

- 0.5 points for explaining the correct effects on C_1 and C_2 .
- 0.5 points for explaining the correct effects on C_3 .
- 0.5 points for explaining the correct effects on SSE.
- -0.5 points for correct explanation without explicitly specifying C_1 , C_2 , and C_3 (or missing any of them).
- No further partial credit.

(b) (1.5 point)

Solution: Three methods: (only one is enough)

1. Remove data points that are much farther from the centroids than other data points **during the K-Means clustering process**. To avoid misidentifying non-outliers and ensure clustering stability, **we must monitor these possible outliers over a few iterations before deciding to remove them**.

2. Perform random sampling. Since sampling selects only a small subset of data points, **the chance of choosing an outlier is very small**. Assign the remaining data points to the clusters using distance or similarity comparisons, or classification.

3. Use KMedian. Compute the median rather than the mean when updating the centroids, as this is more robust to outliers.

Marking scheme:

- 0.5 points for proposing the method.
- 1 point for explaining the method. 0.5 points if a minor mistake or a slightly incomplete explanation. Specific cases:

- -1.5 for incorrectly increasing k .
- -1 for removing outliers without a full explanation.
- -1 for reasonable data processing but not about KMeans itself.
- -0.5 for not monitoring possible outliers over a few iterations before removing them.
- No further partial credit.

5. (1 point)

Solution: *Training:* iterative reassignment and centroid recomputation to fit K clusters. *Prediction:* assigning a *new* point to the nearest learned centroid.

Marking scheme:

- 0.5 points for training.
- 0.5 points for prediction.
- No further partial credit.

6. (1 point)

Solution: Not appropriate: K-Means assumes roughly spherical/convex clusters and equal variance; it cannot capture non-convex shapes like crescents, leading to poor separation.

Marking scheme:

- 0.5 points for answering “No”.
- 0.5 points for correct reason.
- No further partial credit.

Question 7 [10 points]
(Artificial Neuron)

1. (2 points)

Solution: No, a single artificial neuron with sigmoid activation function **can-not** classify this data perfectly.

Justification: Even though sigmoid function is not linear, when used in a single artificial neuron, the decision boundary is $\sigma(z) = 0.5$, which is $z = 0$, which is exactly $w_1x + w_2y + \theta = 0$, so it still produces a linear decision boundary. The given data requires a circular decision boundary to separate the inner disk from the outer ring. Since no straight line can perfectly enclose one circular region while excluding the other (as any line will either cut through the gap region incorrectly or fail to enclose the circular region), the data is not linearly separable.

Marking scheme:

- 0.5 points for answering “No”.
- 1 point for justifying a single artificial neuron with sigmoid activation function produces a linear decision boundary.
 - If only stating the decision boundary is linear but without deriving the decision boundary from the sigmoid function, 0.5 points.
- 0.5 points for stating the data is not linearly separable.
- No further partial credit.

2. (a) (6 points)

Solution:

x_1	x_2	T	$z = w_1x_1 + w_2x_2 + \theta$	y	w_1	w_2	θ
–	–	–	–	–	0.2	-0.3	0.1
2	1	1	$0.2 \times 2 + (-0.3) \times 1 + 0.1 = 0.2$	1	0.2	-0.3	0.1
1	2	1	$0.2 \times 1 + (-0.3) \times 2 + 0.1 = -0.3$	-1	0.4	0.1	0.3
-1	-1	-1	$0.4 \times (-1) + 0.1 \times (-1) + 0.3 = -0.2$	-1	0.4	0.1	0.3
2	1	1	$0.4 \times 2 + 0.1 \times 1 + 0.3 = 1.2$	1	0.4	0.1	0.3
1	2	1	$0.4 \times 1 + 0.1 \times 2 + 0.3 = 0.9$	1	0.4	0.1	0.3
-1	-1	-1	$0.4 \times (-1) + 0.1 \times (-1) + 0.3 = -0.2$	-1	0.4	0.1	0.3

Marking scheme:

0.2 points for each correct entry. No further partial credit.

(b) (1 point)

Solution: Yes, it converges. The decision boundary: $4x_1 + x_2 + 3 = 0$.**Marking scheme:**

- 1 point for a correct decision boundary (not unique, the parameters can be scaled).
- If only answering “Yes”, but wrong/no decision boundary, 0.5 points.
- No further partial credit.

(c) (1 point)

Solution: Classifying the point $(1, 0)$ with final weights $(w_1, w_2, \theta) = (0.4, 0.1, 0.3)$:

$$0.4 \times 1 + 0.1 \times 0 + 0.3 = 0.7$$

$$y = 1 \quad (\text{since } 0.7 > 0)$$

The point $(1, 0)$ is classified as 1.**Marking scheme:**

- 1 point for correct classification.
- If the former calculation is wrong but the classification is correct, still get 1 point.
- If the final classification is wrong, but the weighted sum is correct (which = 0.7), 0.5 points.
- No further partial credit.

Question 8 [10 points]

(Perceptron and Multilayer Perceptron)

1. (2 points)

Solution: Yes. One possible choice: $w_1 = 1$, $w_2 = 1$, $\theta = -0.5$.**Marking scheme:**

- 2 points for a correct set of parameters (not unique, needs manual verification when grading).
- If only answering “Yes”, but wrong/no parameters, 1 point.
- If only answering 3 parameters but wrong, 1 point for “Yes”.
- No further partial credit.

2. (2 points)

Solution: A single perceptron cannot classify this dataset, since it is not linearly separable.**Marking scheme:**

- 1 point for answering “No”.
- 1 point for correct justification.
- Mentioning “not linearly separable”, “cannot find a line to separate”, “XOR question”, or something similar gains full marks.
- No further partial credit.

3. (4 points)

Solution: Yes. One valid parameter set is:

$$\begin{aligned}w_1 &= 1, w_2 = 1, \theta_1 = -1.5; \\w_3 &= -1, w_4 = -1, \theta_2 = 0.5; \\w_5 &= 1, w_6 = 1, \theta_3 = -0.5.\end{aligned}$$

Marking scheme:

- 4 points for a correct set of parameters (not unique, needs manual verification when grading).
- If only answering “Yes”, but wrong/no parameters, 1 point.
- If only answering 9 parameters but wrong, 1 point for “Yes”.
- No further partial credit.

4. (a) (1 point)

Solution: Non-linear activations enable the network to learn complex, non-linear mappings.

Marking scheme:

1 point for correct explanation. If the answers mention that linear activation makes MLP models equivalent to a single perceptron or not better than a single perceptron, 1 point.

No partial credit.

(b) (1 point)

Solution: Without them, the MLP collapses into an equivalent single linear transformation, offering no improvement over a simple perceptron.

Marking scheme:

1 point for correct explanation. If the answers mention linear combination, 1 point. No partial credit.